

Multi-User Mixed Reality Control of a Multi-Arm Robotic System

Saul Alexis Heredia Perez ^{*}
The University of Tokyo

Hung-Ching Lin [†]
The University of Tokyo

Kanako Harada [‡]
The University of Tokyo



Figure 1: Multi-arm robot system (left), its Digital Twin (center), and the proposed Mixed Reality user interface for its control (right).

ABSTRACT

This work presents a mixed reality (MR) user interface integrated with a Digital Twin (DT) for collaborative control of a multi-arm robotic system in laboratory environments. The DT maintains a synchronized virtual representation of the robots and task scene, enabling multiple operators to interact in a shared MR workspace. Using HMDs and motion tracking, users intuitively select and control robot arms through natural hand motions. A preliminary user study with a collaborative manipulation task demonstrates reduced task completion time and lower subjective workload compared to a conventional leader–follower interface, highlighting the system’s potential for intuitive multi-user robot teleoperation.

Index Terms: Human-machine interaction, Digital Twin, Collaborative virtual tasks

1 INTRODUCTION

Robotic manipulation of sterilized materials is increasingly important in biomedical and laboratory environments, where tasks such as pipetting, weighing, and handling clean instruments require precision while limiting direct human intervention. Collaborative robotic systems offer a promising path toward automating these processes, but collecting high-quality demonstrations remains challenging, particularly in clean or hazardous settings where physical interaction with robots is impractical [2].

Leader–follower teleoperation provides a practical alternative to direct teaching. However, conventional interfaces such as bimanual haptic devices and video-based control are primarily designed for single operators and do not scale well to multi-arm, multi-user scenarios [5]. Rigid mappings between users and robots, reliance on direct visibility, and increased cognitive load make collaborative manipulation difficult, especially for precision laboratory tasks performed under sterilization constraints.

Virtual reality (VR) simulators provide safe environments for training and task planning but lack real-time synchronization with physical systems. Digital Twins (DTs) address this limitation by continuously updating virtual representations of robots using live

sensor data, enabling interactive monitoring and control. Mixed Reality (MR) further integrates physical and virtual environments, allowing operators to interact with robots within a shared spatial frame. Despite increasing interest in MR for robot interaction, most prior work focuses on single-user interfaces or isolated tasks, and collaborative multi-user control of high-dimensional, multi-arm robotic systems remains underexplored [3]. While Davari et al. [1] proposed Augmented Reality (AR) interfaces for human–robot collaboration limited to single-user, single-robot scenarios, Szczurek et al. [6] introduced a multi-user MR interface without evaluating operator workload or comparing between different interfaces.

To address this gap, this work presents a MR interface integrated with a DT that enables multiple operators to collaboratively control a four-arm robotic system in a shared virtual space. The system architecture and a preliminary user evaluation are presented, demonstrating improved task performance and reduced workload compared to conventional leader–follower control.

2 SYSTEM OVERVIEW

The proposed system integrates a physical multi-arm robotic platform with four Franka Research 3 (Franka Robotics, Germany) 7-DOF robot arms equipped with a Franka Hand (Franka Robotics, Germany) as the end effector, a Digital Twin (DT), and a Mixed Reality (MR) user interface to enable real-time collaborative multi-user teleoperation (Fig. 1). A preliminary version of this system without user evaluation was presented in [4]. The DT maintains a shared virtual representation of the robot system and workspace, while a server component synchronizes robot state and user inputs to ensure a consistent scene across all clients.

Operators interact using a Meta Quest 3 (Meta Platforms, Inc., USA) head-mounted display, which provides an immersive shared workspace for selecting and controlling robot arms through hand-based interaction. Spatial alignment between the virtual and physical environments is achieved using an OptiTrack (NaturalPoint, Inc., USA) motion-tracking system, establishing a unified coordinate frame shared by all users. Robot motion is generated through an inverse kinematics controller augmented with a VFI-based safety layer enforcing joint limits and collision avoidance via simple geometric constraints. The robot end-effector follows the user’s hand pose in 6 DOF only while an explicit control button on the controller is pressed with visual feedback indicating when robot control is active, and a separate button controls gripper actuation. The end-to-end system latency, including DT synchronization, MR rendering, and robot response, is 22.7 ms.

^{*}e-mail: saulheredia@g.ecc.u-tokyo.ac.jp

[†]e-mail: qlin1806@g.ecc.u-tokyo.ac.jp

[‡]e-mail: kanakoharada@g.ecc.u-tokyo.ac.jp

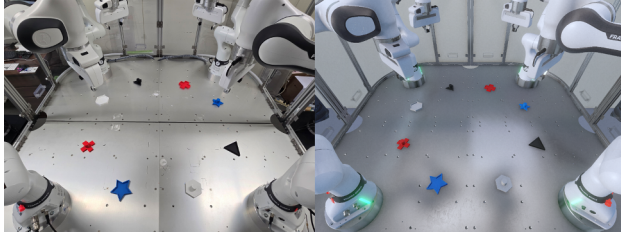


Figure 2: Task setup in real life (left) and in its DT (right).

3 APPLICATION SCENARIO AND TASK

The system is demonstrated using a collaborative peg-in-hole task as a surrogate for laboratory automation scenarios requiring precise, coordinated manipulation under restricted access conditions. Four pegs, each consisting of a 30 mm cylindrical handle attached to a 100 mm block with a distinct geometric shape, must be inserted into matching holes with a 2 mm tolerance (Fig. 2). Two operators work simultaneously, with one transferring a peg and the other performing the insertion. Failed insertion attempts can be retried until all pegs are placed, constituting a single trial. The spatial arrangement enforces multi-arm collaboration and verbal coordination. Due to the lack of physical object tracking, peg transfer and insertion are simulated within the DT, while robot motion is executed on the physical system. The DT is spatially aligned and presented at the same height for all participants.

4 PRELIMINARY EVALUATION

A preliminary user study compared the proposed MR interface with a conventional leader–follower setup using stylus-type haptic devices (Touch X, 3D Systems, USA), one per robot (Fig. 3). Six participants worked in pairs in a within-subjects design, completing one trial per user interface in counter-balanced order. Task completion time was measured from the first peg grasp to the last insertion, and subjective workload was assessed using the unweighted NASA-TLX after each condition. IRB approval was not required for this pilot study. Statistical significance was assessed using an paired, two-tailed *t*-test. The MR interface enabled significantly faster task completion ($M = 93.8$ s, $SD = 6.2$) than the conventional method ($M = 220.0$ s, $SD = 58.9$; $p < 0.005$) and exhibited lower performance variability. Subjective workload was also reduced, with a significantly lower total NASA-TLX score for the MR interface ($M = 34.2$, $SD = 8.7$) compared to the conventional interface ($M = 58.3$, $SD = 23.4$; $p < 0.05$) (Fig. 4).

5 DISCUSSION AND LIMITATIONS

The study indicates that a MR interface integrated with a DT can support real-time, collaborative control of a multi-arm robotic system. A unified spatial calibration enabled all users to share a consistent workspace, facilitating safe and intuitive collaboration. In the evaluated task, the MR interface achieved shorter completion times and lower subjective workload than a conventional leader–follower setup, suggesting improved usability for collaborative manipulation. Limitations include the small participant sample, a simplified task, and the absence of physical object tracking, with task objects represented only in the DT. Despite these constraints, the results demonstrate the feasibility of the proposed interaction approach.

6 CONCLUSION AND FUTURE WORK

This work demonstrates a MR human–robot interface integrated with a DT for intuitive, multi-user control of a multi-arm robotic system. The system highlights the potential of shared MR environments to improve efficiency and reduce operator workload in

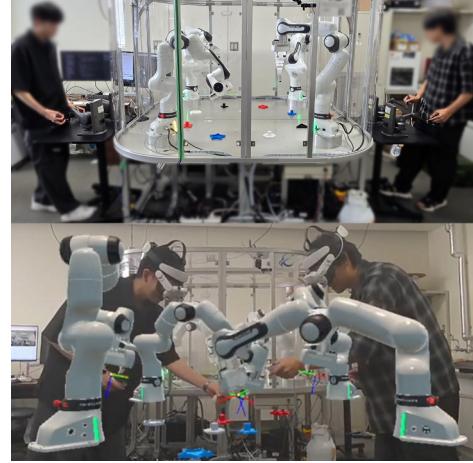


Figure 3: Operating the robot using a conventional user interface (top) and the proposed MR-DT user interface (bottom).

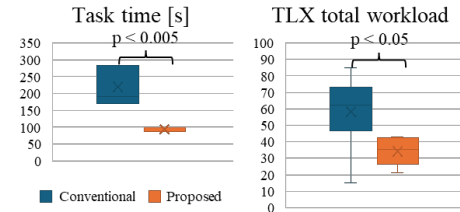


Figure 4: Results from task completion time and NASA-TLX.

collaborative robot manipulation. Future work will focus on integrating physical object tracking, expanding to more complex laboratory tasks, and conducting larger-scale user studies to further evaluate the system’s applicability to collaborative robot operation and demonstration data collection.

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